

ASSIGNMENT – I

PLAIN SCALES

1. An area of 49 square centimeters on a map represents an area of 16 sq. m on a field. Draw a scale long enough to measure 8 m. Mark a distance of 6 m 9 dm on the scale.
2. A cube of 5 cm side represents a tank of 1000 cubic meters volume. Find the R.F and construct a scale to measure up to 35 m. Mark a distance of 27 m on it.

DIAGONAL SCALES

1. The distance between two stations by road is 200 km and it is represented on a certain map by a 5 cm long line. Find the R.F and construct a diagonal scale showing a single kilometer and long enough to measure up to 600 km. Show a distance of 467 km on this scale.
2. An area of 400 square centimeters on a map represents an area of 25 square kilometers on a field. Construct a scale to measure up to 5 km and capable to show a distance of 3.56 km. Indicate this distance on the scale.

CONIC SECTIONS

1. A fixed point is 90 mm from a fixed straight line. Draw the locus of a point P moving in such a way that its distance from the fixed straight line is twice its distance from the fixed point. Name the curve. Draw a tangent and a normal at point 40 mm away from the fixed point.
2. Draw the locus of a point which moves in such a manner that its distance from a fixed point is equal to its distance from a fixed straight line. Consider the distance between the fixed point and the fixed line as 60 mm. Name the curve.
3. The focus of a hyperbola is 60 mm from its directrix. Draw the curve when eccentricity is $\frac{5}{3}$. Draw a tangent and a normal to the curve at a point 45 mm from the directrix.

CYCLOIDS

1. ABC is an equilateral triangle of side 50 mm. Trace the loci of vertices A, B and C when the circle circumscribing triangle ABC rolls without slipping along a fixed straight line for one revolution.
2. Draw the locus of a point lying on the circumference of a wheel of diameter 60 cm for one revolution, when it passes over a segmental arched culvert of radius 1 m. Take a scale of 1:10
3. A circus man rides on a motorcycle inside a globe 3.5 m diameter. The motor cycle wheel is 1 m in diameter. Draw the locus of a point lying on the circumference of the wheel of the motorcycle for one turn. Take a scale of 1:20.

Assignment – II

PROJECTIONS OF POINTS

1. Draw the projections of the following points on a common reference line keeping the distance between their projectors 25 mm apart.
 - (a) Point A is 40 mm above the H.P. and 25 mm in front of the V.P.
 - (b) Point B is 40 mm above the H.P. and in the V.P.
 - (c) Point C is 25 mm in front of the V.P. and in the H.P.
 - (d) Point D is 25 mm above the H.P. and 30 mm behind the V.P.
 - (e) Point E is in the H.P. and 30 mm behind the V.P.
 - (f) Point F is 40 mm below the H.P. and 30 mm behind the V.P.
 - (g) Point G is 25 mm below the H.P. and 40 mm in front of the V.P.
 - (h) Point H is in the V.P. and 30 mm below the H.P.

PROJECTIONS OF STRAIGHT LINES

1. A 100 mm long line PQ has its end P 10 mm above the H.P. and 70 mm in front of the V.P. The line is inclined at 60° to the H.P. and 30° to the V.P. Draw its projections
2. A line measuring 75mm long has one of its ends 50 mm in front of VP and 15 mm above HP. The top view of the line measures 50mm. The other end is 15mm in front of VP and above HP. Draw and measure the front view. Determine the true inclinations.
3. A line PQ 85 mm long has its end P 10 mm above HP and 15 mm in front of VP. The top view and front view of line PQ are 75 mm and 65 mm respectively. Draw its projections. Also determine the true and apparent inclinations of the line.
4. The front and top views of the 75 mm long line PQ measures 50 mm and 60 mm, respectively. If the end P of the line is 35 mm above the H.P. and 15 in front of the V.P., draw its projections and locate the traces. Determine the true inclinations of the line PQ with the H.P. and the V.P.
5. The distance between the end projectors of a line PQ is 50 mm. The end P is 50 mm in front of the V.P. and 25 mm above the H.P. The end Q is 10 mm in front of the V.P. and above the H.P. The line is inclined at 30° to the V.P. Draw the projections of line PQ. Determine its true length and true angle of inclination with the H.P.

PROJECTIONS OF PLANES

1. A circular plane of diameter 50 mm is resting on a point of the circumference on the H.P. The plane is inclined at 30° to the H.P. and its centre is 35 mm in front of the V.P. Draw its projections.
2. A rectangular plane of sides 70 mm and 35 mm has a shorter side on the H.P. The surface of the plane is inclined at 60° to the H.P. and perpendicular to the V.P. Draw its projections.
3. A rhombus of diagonals 70 mm and 45 mm is placed on an end of the major diagonal on the H.P. Its surface is inclined to the H.P. and perpendicular to the

V.P. such that the top view appears as a square. Draw its projections and determine inclination of the rhombus with the H.P.

4. A trapezium ABCD having parallel sides AB = 50 mm, CD = 30 mm and height 40 mm is kept on its side AB in the V.P. The surface of the trapezium is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections.
5. The top view of a triangular plane is an equilateral triangle of side 50 mm having one side parallel to the reference line. The front view of the plane is a 65 mm long straight line. Determine the true shape of the plane. Surface inclined to V.P. and perpendicular to H.P.
6. A hexagonal lamina of side 50 mm has a corner in the V.P. The diagonal through that corner is parallel to H.P. and inclined at 30° to the V.P. Draw its projections when the lamina is perpendicular to the H.P. Measure the distance of the topmost corner from the V.P.
7. A pentagonal plane of side 30 mm rests on an edge in the V.P. with its surface perpendicular to the H.P. The corner opposite to the edge on which it is resting, is 30 mm in front of the V.P. Draw the projections of the plane and determine its inclination with the V.P.
8. A semi-circular plane of diameter 70 mm has its straight edge on the V.P. and is inclined at 30° to the H.P. Draw the projection of the plane when its surface is inclined at 45° to the V.P.